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Key Points:

- An immediate ice stream response to calving was observed up to 11 km from the terminus of a major Greenland outlet glacier
- Enhanced deformation rates were instantaneously observed in the highly-crevassed zone of the ice stream shear margins
- The loss of lateral drag due to calving largely contributed to such a widespread calving response

Supporting Information:

Supporting Information may be found in the online version of this article.

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Immediate Ice Stream and Shear Margin Response to Calving at Sermeq Kujalleq in Kangia, Greenland

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Abstract The sensitivity of Greenland outlet glaciers to short-term flow disturbances lacks observations to be well constrained in numerical models. Specifically, sudden gigaton-scale calving events regularly occurring at major Greenland outlet glaciers like Sermeq Kujalleq in Kangia (SKK; also known as Jakobshavn Isbræ) are known to perturb background ice flow in complex ways. Here, we present high-rate terrestrial radar observations and show an immediate step-wise acceleration of SKK ice stream up to 11 km from its terminus, one of the longest immediate calving responses so far reported in Greenland. Focusing on SKK's shear margins, we detected large instantaneous increases in deformation rates in its most crevassed section, which, together with the along-flow response, provide evidence for strong lateral and longitudinal coupling of the ice stream. Using a simplified theoretical framework, we show that the loss of lateral drag due to calving is a key component of such a widespread calving response.

Plain Language Summary The sudden rupture of deep crevasses at marine-terminating Greenland glaciers releases large ice blocks to the ocean that can weigh more than a gigaton. This process, called iceberg calving, is a key component of the mass loss of the Greenland ice sheet, and it can in turn impact glacier speeds over durations much longer than the brief lifespan of the process itself. Yet, the processes and conditions linking calving events with resulting velocity changes still remain unclear. Here, we study the surface velocity of Sermeq Kujalleq in Kangia (SKK; also known as Jakobshavn Isbræ), Greenland's fastest ice stream, during a large calving event using field observations with high temporal (minutes) and spatial (meters) resolution. We show that the calving event resulted in an immediate acceleration of the ice stream up to 11 km from its terminus. We also observed large increases in deformation rates at the highly crevassed transition between fast stream flow and slower glaciated surroundings, called the shear margin. Using a simplified numerical model, we show that the loss of resistive force that the sides of the ice block was providing to the ice stream was a major contributor to this acceleration.

1. Introduction

The flow acceleration and enhanced discharge of tidewater glaciers in Greenland (Kalaallit Nunaat) initiated two decades ago accounted for half of the total mass loss of the Greenland ice sheet over this period (Shepherd et al., 2020). While the multi-annual and seasonal variations in surface velocity of Greenland glaciers have been widely studied using spaceborne observations (e.g., Black & Joughin, 2022; Howat et al., 2007; Joughin et al., 2004) and numerical models (e.g., Aschwanden et al., 2016; Morlighem et al., 2016; Nick et al., 2009), much less is known about fluctuations at daily to hourly scales and their influence on longer-term ice flow. Among the processes acting over such short periods, iceberg calving plays a key role in the mass loss of the Greenland ice sheet (Mankoff et al., 2020), but still remains poorly constrained in numerical models owing to fast-changing and complex conditions at the ice-ocean boundary.

A wide variety of velocity responses have been observed at the front of Greenland tidewater glaciers following calving events (e.g., Amundson et al., 2008; Holland et al., 2016; Murray et al., 2015; Podrasky et al., 2012; Sundal et al., 2013) with complex links between iceberg size and conditions at the terminus (Cassotto et al., 2019). At Sermeq Kujalleq in Kangia (SKK; also known as Jakobshavn Isbræ), Greenland's fastest ice stream (Moon et al., 2014) with surface speeds up to 45 m d⁻¹, calving-driven speedups have been observed over distances ranging 1–4 km from the glacier terminus (Amundson et al., 2008; Podrasky et al., 2014; Rosenau et al., 2013).

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Although modeling results suggested calving responses can be visible further than 5 km upstream (Amundson et al., 2022), nearly-instantaneous calving responses exceeding such distances have so far only been reported once. At Helheim glacier, Nettles et al. (2008) observed a speedup immediately after calving at the uppermost point of their global navigation satellite system deployment, 20 km from the terminus. The glacier conditions and physical processes behind this extraordinary response distance still remain unresolved.

Studies targeting calving responses so far mainly focused on along-flow velocity variations, from frontal to upstream areas. Fast-flowing Greenland outlet glaciers exhibit large areas of slower-moving ice in the surroundings of their main stream creating highly-crevassed transitional zones in the form of shear margins. At SKK specifically, shear margins have been identified as key areas in the dynamics of the ice stream (Van Der Veen et al., 2011). Despite their importance, the study of their short-term response to abrupt forcings was only initiated recently (e.g., Wehrlé et al., 2024) and lacks both observational and theoretical considerations.

Here, we analyze observations of a stepwise ice-flow acceleration immediately following a calving event at the front of SKK in July 2023 using a terrestrial radar interferometer (TRI). We present velocity variations along the flow of the ice stream to determine the acceleration amplitude and propagation length. We further investigate the response of the shear margins to this calving speedup through an across-flow profile at a distance of 3 km from the terminus. This calving event and associated velocity response are then used as a full-scale natural experiment and are analyzed in a simplified theoretical framework to better understand the properties and sensitivity of the ice stream to short-term forcings.

2. Methods

A terrestrial radar interferometer (TRI) system (Caduff et al., 2014; Werner et al., 2008) was deployed during 13 days in July 2023 (July 5th–17th) at the front of SKK to observe the ice stream and calving dynamics. The installation site on bedrock is 334 m.a.s.l at a distance of 5.5 km from the calving front (Figure 1a) and acquired data up to a range distance of 20 km (Figure 1b). The TRI system was a Gamma Portable Radar Interferometer operating at a wavelength of $\lambda = 17.4$ mm (Ku-Band, 17.2 GHz) and was scanning the area of interest every two minutes, enabling the detection of sub-centimeter scale displacements. The range resolution is approximately 0.75 m while the azimuth resolution is 0.4° , corresponding to 31.4 m at a slant range of 4.5 km. An azimuth step of 0.1° was used, corresponding to a sampling interval of 7.8 m in azimuth at a similar slant range.

Velocity maps in line of sight (LOS; i.e. surface displacements in the TRI viewing direction) were computed by unwrapping 2-min spaced interferograms (Werner et al., 2008) from reference points on slow-moving glaciated areas directly adjacent to the ice stream. LOS velocities were then projected onto observed flow lines to obtain absolute velocities while conserving short-term and local velocity variations (see Text S1 in Supporting Information S1).

Flow velocities were extracted on a profile from the vicinity of the terminus up to 11 km upstream (point A1 to A2 on Figure 1a) to study along-flow velocity changes following the calving event. At 3 km upstream from the terminus, the transition from slow-moving ice to fast ice stream flow (point B1–B2 on Figure 1a), referred to as shear margin, could also be extracted without shadow effects that are otherwise common in the radar signal across the ice stream margins (see Figure 1b). As atmospheric noise increases with distance from the TRI system while surface velocities decrease, the velocity extracted along both profiles was then averaged over an hour before (07–15 00:00–01:00) and after (07–15 02:00–03:00) the calving event to improve signal-to-noise ratios. Two-minute spaced time series were also extracted at three discrete points 3, 6, and 11 km along profile A1–A2 (points C, D and A2 on Figure 1a, respectively). Calving masses were computed by manually delineating calving areas on TRI backscatter images (see e.g. Figures 1c and 1d) assuming an ice thickness of 900 m at the terminus (a rough estimate; Lüthi et al., 2009), and an ice density of 917 kg.m^{-3} .

3. Results

Surface velocities extracted at three locations along the main trunk of SKK ice stream (3, 6 and 11 km from the terminus) show respective velocity increases of 23%, 16%, and 10% between the beginning and the end of the 13-day study period (Figures 2a–2c). Seven large calving events (considered as >0.2 Gt) were observed during the study period (vertical dashed lines on Figures 2a–2c) and resulted in a variety of flow responses that we suggest is an illustration of a non-linear response to calving events observed in the past at SKK (Cassotto et al., 2019).

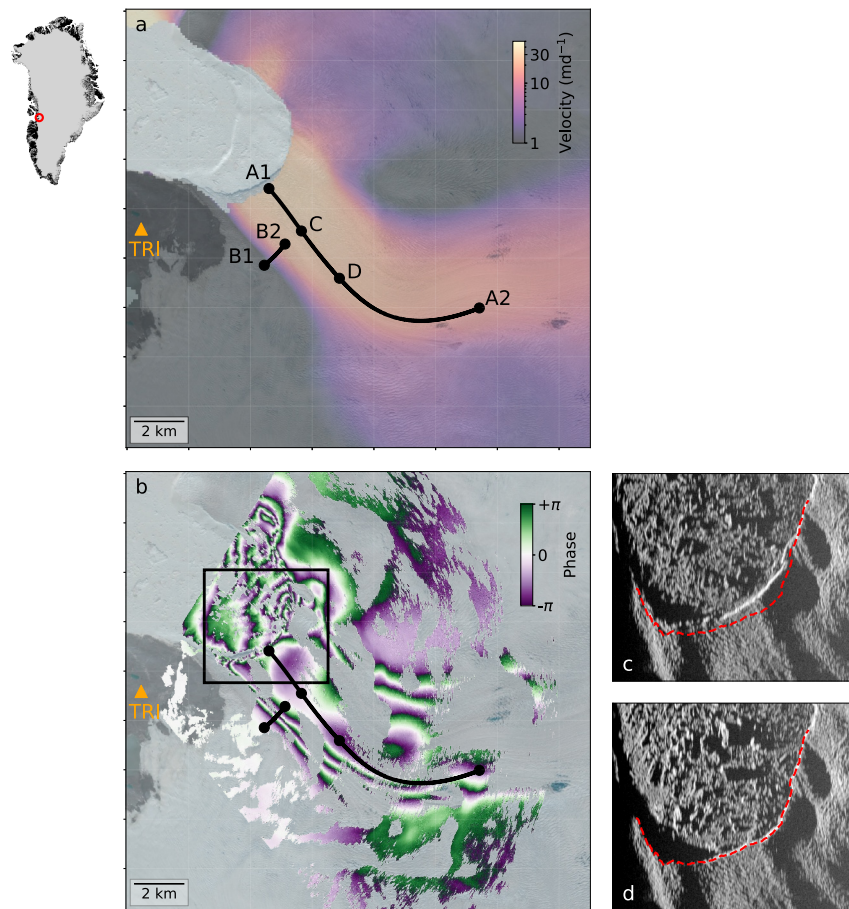


Figure 1. (a) July 2023 mean velocity (Howat et al., 2022) overlaying a Sentinel-2 enhanced true color image of SKK on 12/07/2023 (ESA, 2023). Along-flow profile A1–A2 and shear margin profile B1–B2 are indicated by black lines, and points C and D together with profile ends, by black dots. (b) 10-min stacked terrestrial radar interferometer (TRI) interferogram over the study area. 2-minute spaced TRI backscatter images before (c) and after the calving event (d). The red dashed line represents the terminus position after the calving event.

At 01:30 on July 15th, a large calving event was observed where a ~ 160 m-long block of ice broke off across the full terminus width of ~ 5 km (Figures 1c and 1d). This calving event was the second largest recorded during the field campaign with a mass of ~ 0.7 Gt. High-rate TRI backscatter imagery showed that it originated from a large rift opening that propagated from south to north across the terminus. Two major break-off events closely following each other could be identified during the propagation of the rift rupture (red dashed lines on Figures 2d and 2e).

This large calving event caused a sudden acceleration during which the velocity monitored 3 km from the terminus increased by 7.1% in less than 10 min, before stabilizing at 3.1% above the pre-event velocity within half an hour (Figure 2e). This acceleration was two to three orders of magnitude higher than the longer-term acceleration observed during the 13-day field campaign. At this location, the immediate flow response resulted in an additional horizontal displacement of 0.04 m within the first 30 min following the calving event, and amounted to 0.10 m two hours after the event (Figure 2d). The velocity increase was also immediately visible along the SKK ice stream 6 and 11 km from the terminus (Figures 2b and 2c), and corresponded on average to 16.5% of the longer-term increase between the beginning and the end of the field campaign.

The analysis was further extended to velocity changes over one hour before and after the calving event on the along-flow profile A1–A2, and revealed a 4.5% acceleration with respect to pre-event velocity in the vicinity of the terminus (Figure 2f). The damping of the acceleration with distance from the terminus could be approximated with a linear decay of $-0.4\%/km$ upstream of the first 800 m from the terminus. Despite the signal quality

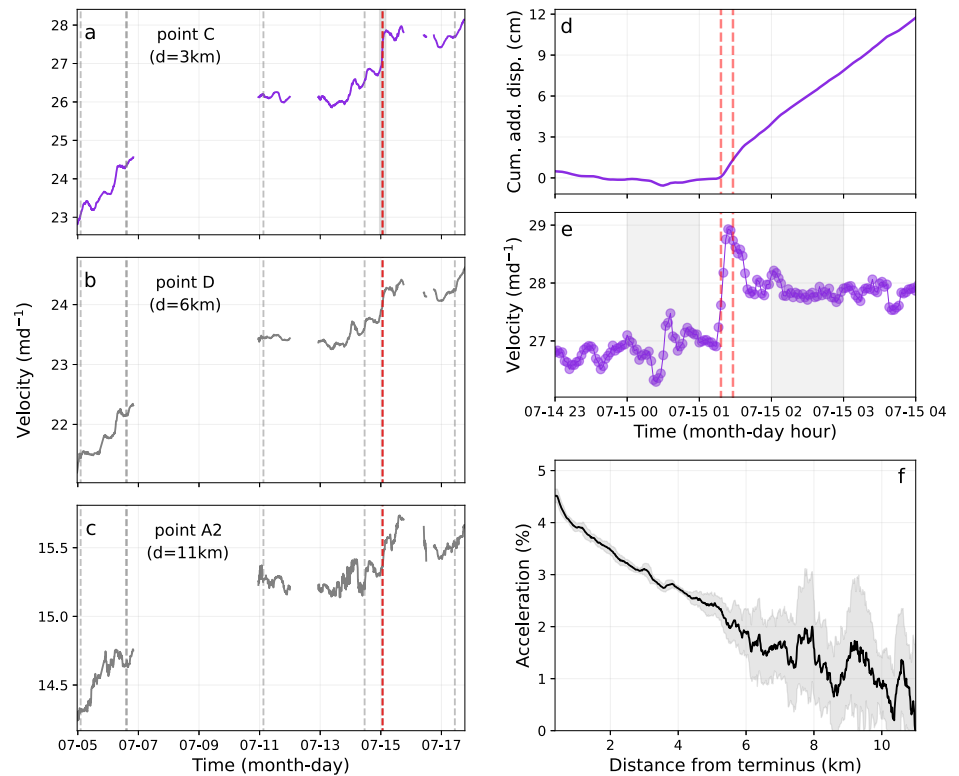


Figure 2. (a, b, c) Surface velocity during the study period at points C, D and A2 that are at a distance of 3, 6 and 11 km from the terminus, respectively. Data gaps correspond to periods of strong wind during which the terrestrial radar interferometer was taken down. (d) Cumulative additional displacement (see Text S1 in Supporting Information S1) at point C following calving event. (e) Surface velocity at point C. (f) Along-flow acceleration on the profile A1–A2 over one hour before and after calving event (shown as gray areas in panel e). Calving event timings are shown as gray dashed lines, red dashed lines highlighting the calving events of interest. A gray envelope representing $\pm 1\sigma$ on the rolling mean (see Text S1 in Supporting Information S1) in (f) is shown to illustrate signal degradation with distance from the instrument.

deteriorating farther than 5 km from the terminus, an acceleration was still clearly visible up to 11 km. Since the discrete point analysis at this location, which includes a more localized treatment of atmospheric noise (see Text S1 in Supporting Information S1), indicated a 1.6% acceleration relative to the pre-event velocity (Figure 2c), this suggests that the total response length exceeded 11 km.

At 3 km from the terminus and before the calving event, the main trunk velocity of 26.2 md^{-1} (point B2 on Figure 1a) decreased toward the shear margins (orange curve on Figure 3a), with the velocity gradient reaching a maximum midway along profile B1–B2 before weakening toward the outer shear margin, where the velocity approached zero (point B1 on Figure 1a). After the calving event, an immediate acceleration was observed across the shear margin. The velocity in the main trunk increased by 0.87 md^{-1} (black curve on Figure 3a), that is 3.3% above pre-event velocities, in close accordance with nearby along-flow changes (Figures 2e and 2f). From the main trunk, the acceleration signal rapidly decayed through the shear margin and became undetectable beyond 1 km ($x = 0.4 \text{ km}$; see Figure 3a).

The shear rate increase after the calving event, that is the magnitude of the change in shear rate, featured an almost symmetric shape with a peak at $2.5 \cdot 10^{-3} \text{ d}^{-1}$ in the middle of the shear margins ($x = 0.74 \text{ km}$; purple dashed line on Figure 3b). In the inner and outer areas surrounding this peak, the shear rate change plateaued at half the peak amplitude for 150 m. Changes in the outermost section of the shear margin ($x = 0\text{--}0.3 \text{ km}$) and toward the main trunk ($x = 1.1\text{--}1.4 \text{ km}$) were strongly reduced, yet the main trunk still sustained values above $0.3 \cdot 10^{-3} \text{ d}^{-1}$ before approaching null changes at the end of the profile. The shape of the shear rate changes induced by calving is strikingly similar to the one observed during the 13 days of the study period (gray curve on Figure 3b). The peak shear rate increase following the calving event amounts to 18% of the peak increase over the study period, and the total increase across the profile, to 20%. Although low signal quality precluded a detailed analysis of the shear

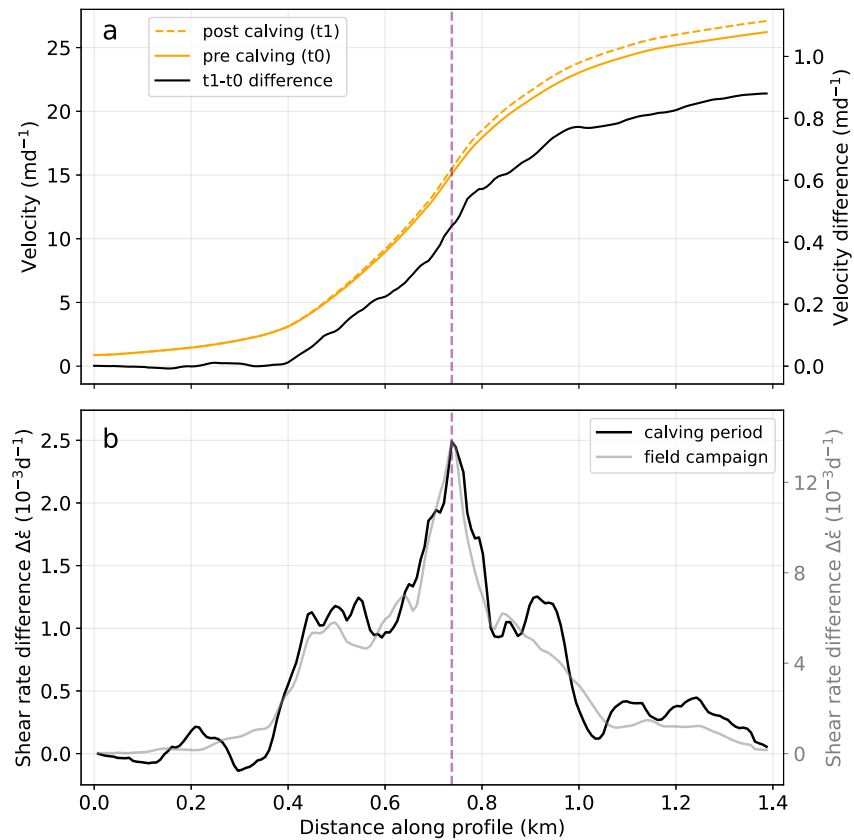


Figure 3. (a) Surface velocity across the shear margins from slow-moving ice to the main trunk (B1–B2 on Figure 1a) average over one hour before and after calving event, and velocity differences over the same period. (b) Shear rate differences (see Text S1 in Supporting Information S1) over the calving period (black) and the field campaign (gray). The red dashed line shows the area of maximum shear rate difference.

margin response to calving further upstream, an immediate acceleration following calving was also observed on a profile 7 km upstream from the terminus, up to 3 km into the shear margin (Figure S1 in Supporting Information S1). This suggests a widespread response of the shear margins along the ice stream.

4. Discussion

4.1. Ice Stream Coupling and Sensitivity to Calving

Within one hour following a large calving event at SKK, a step-wise acceleration of the ice stream was observed up to 11 km upstream of the terminus with a TRI system. The only similar documented event, from Helheim glacier (Nettles et al., 2008), showed an acceleration following calving that peaked at 6% at the glacier terminus and was detectable up to 20 km upstream. While both the propagation length and magnitude are higher than reported here, comparisons remain difficult as no calving event area nor volume could be estimated at Helheim glacier and only indirect indications of a large calving event were obtained using seismics. Differences in glacier and fjord geometries as well as basal conditions between Helheim glacier and SKK would also likely lead to discrepancies. Surprisingly, a non-linear decay with increasing distance from the terminus was observed at Helheim glacier where accelerations down to 1% persisted upstream of 12 km. This suggests, together with the 1.6% acceleration detected 11 km from the terminus of SKK (Figure 2c), that the acceleration reported at SKK may have persisted at low amplitude beyond 11 km.

Across the shear margin of SKK at 3 km upstream of the terminus, an instantaneous acceleration was observed without time lag between the fast main trunk and the slow-moving surrounding ice. The increase in shear rate occurring within one hour following the calving event amounted to 20% of the change observed in 13 days. This suggests that short-term perturbations can lead to persistent dynamic changes. Beyond the amplitude of the shear

rate increase, the distribution of these changes across the shear margin immediately after the calving event is similar to those over the 13-day study period. This suggests that deformations initiated in the main trunk are distributed across the shear margins in a similar way for sudden forcings and multi-day variations. This, however, does not mean that the deformation at a given location is of similar nature for both forcing time scales. Considering rates of shear rate changes, our results reveal that shear rate changes occurred more than an order of magnitude faster due to the sudden calving-induced speedup than during the multi-day glacier acceleration. Such a difference implies different impacts and potentially different types of deformation. Interestingly, the area of highest shear rate changes in the shear margin corresponded to a highly-crevassed zone that expands down to the glacier terminus. The coincidence of maximum shear rate increase with crevassing patterns hence questions whether the observed displacement involved brittle deformation. This also hints toward a potential contribution of calving to further crevasse formation and opening which has, to our knowledge, so far never been documented.

The immediate along-flow and across-flow response to this calving event suggests strong coupling between the frontal and upstream areas of SKK, and between the main trunk and the shear margins. Wehrlé et al. (2024) drew similar conclusions on the coupling and sensitivity of SKK ice stream to sudden disturbances by focusing on its response to a subglacial flood following the drainage of supraglacial lakes, another type of short-term forcing. An immediate response of the shear margins was also observed 7 km from the terminus as reported here (Figure S1 in Supporting Information S1), and over a similar distance through the shear margin (3 and 3.4 km, respectively). This suggests that the shear margins may respond in a similar sensitive manner to different types of short-term forcings affecting the main trunk.

4.2. Conditions for Calving Response

Our observations suggest a strong coupling between the terminus and upstream areas as well as an instantaneous reaction of the shear margins which suggests that lateral drag played an important role in this change of dynamics. However, important questions remain concerning the type of deformation and physical conditions causing this acceleration.

Due to the very high-rate nature of the observed displacements, for which the ice rheology is poorly constrained, it is difficult to assess whether their origin is brittle, viscous, or a combination thereof. Motivated by the long-distance immediate response revealed by our observations, we therefore propose here a simplified representation of the ice stream based on a force balance to help us identify the parameters driving this acceleration without modeling the latter explicitly. Rather than prescribing an uncertain type of deformation, we apply here a framework inspired by theoretical considerations and constrained using observed displacements to circumvent their characterization. This setup therefore allows us to probe the conditions needed to reproduce our observations based on parameters external to flow, that is basal friction, drag at the shear margins and frontal buttressing stress.

We propose a model based on a rigid block of length L given by the reaction length of 11 km, width $W = 5$ km and thickness $H = 1$ km, moving within a viscous fluid, representing the shear margins. From this representation, a force balance between the drag at the shear margins, the basal friction, the driving stress and the force F_f applied at the front of the glacier by ocean water and ice mélange can be established.

After the calving event, the block has a new velocity U_1 and a new shorter length $L_1 = L_0 - L_c$ where L_c is the length of the ice that calved and L_0 the length of the block before calving when the ice block had a velocity U_0 . It can be demonstrated that the ratio between the new velocity U_1 and the velocity U_0 before calving is given by (see Text S2 in Supporting Information S1)

$$\frac{U_1}{U_0} = \left(\frac{\rho_i g H (\alpha - \mu) + \frac{F_f(H_m, \zeta)}{L_1}}{\rho_i g H (\alpha - \mu) + \frac{F_f(H_m)}{L_0}} \right)^n \quad (1)$$

In Equation 1, α is the surface slope of the block estimated to be 0.028 (Morlighem et al., 2017), μ the friction coefficient at the base, ζ quantifies potential variations in the buttressing force and n is the exponent arising from the lateral drag on the margins. n is taken equal to 3 to represent the lateral drag due to glacier flow following Glen's flow law (Hindmarsh, 2006). Note that Equation 1 has been obtained assuming no change in the properties of the shear margins at the timescale of the acceleration (see Text S2 in Supporting Information S1). This remains

in agreement with the large deformations observed in the shear margins (Figure 3) and a potential brittle contribution, as it only considers the absence of changes in properties immediately after the event and does not exclude a longer-term response. This hypothesis is supported by the modeling of the shear margin velocity profiles presented in Figure 3 (see Text S2 in Supporting Information S1).

We further considered the basal friction coefficient to be of similar magnitude as the surface slope ($\alpha \sim \mu$) to reflect the low basal friction coefficient that has been suggested before at SKK (Shapiro et al., 2016). Together with the assumption of an unchanged buttressing F_f shortly after the event, Equation 1 can be simplified to

$$\frac{U_1}{U_0} = \left(1 + \frac{L_c}{L_1}\right)^n \quad (2)$$

to directly probe the effect of the sudden loss of ice with length L_c involved in the lateral drag. Assuming $n = 3$, the ratio between the newly established velocity U_1 and the pre-calving velocity U_0 is equal to 1.045. This closely matches the acceleration observed at the glacier front (Figure 2f). This theoretical consideration shows that if the friction coefficient μ is very close to the surface slope α and the reaction length is $L_1 = 11$ km, then a loss of lateral drag on a length of 160 m is enough to cause the observed acceleration at the front.

One-dimensional depth-averaged modeling (Text S2 in Supporting Information S1) suggests a basal friction coefficient that is twice lower than assumed above (0.015 and 0.028, respectively). In order to match the observed acceleration with this lower basal friction coefficient, Equation 1 either implies a different buttressing coefficient in the two steady states or a reaction length shorter than 11 km (Text S2 and Figure S5 in Supporting Information S1). A different buttressing effect following the event could be explained by the stiffening and weakening impacts of calving on the ice mélange (Amundson et al., 2010; Wehrle et al., 2023). However, for this change to be significant, a clear and sustained shift in the ice mélange regime immediately after the calving event would be required (Figure S5d in Supporting Information S1), which our TRI data does not suggest. While this difference in friction coefficients may stem from discrepancies in modeling frameworks, both nevertheless retrieve a low basal friction coefficient of a similar order of magnitude.

A low basal friction coefficient therefore appears to be necessary in order to match an acceleration of the same order of magnitude as observed. This conclusion is in agreement with previous studies suggesting that a Coulomb friction can match observed SKK dynamics (Trevers et al., 2024), and provide further evidence for a weak layer with very low basal friction at SKK (Shapiro et al., 2016), at least on the first kilometers from the terminus. This suggests that the drag on the shear margins could be the main process that opposes the extra extensive stress in such abrupt accelerations.

The theoretical framework presented here considered SKK accelerating as a block (uniform and constant acceleration over the coupling length) and could reproduce the acceleration observed at its terminus where the ice is the least coupled to upstream areas. By design, it could not reproduce the observed decreasing acceleration with distance from the terminus, which would require explicit modeling of short-term internal deformations within the ice block, an approach that remains loosely constrained. This model is hence useful to identify potential processes that trigger the acceleration, but the full shape of the along-flow acceleration is still to be explained.

Based on Equation 2, the existence of a reaction length over which deformation occurs appears to be an important parameter to predict the magnitude of the acceleration, and suggests strong coupling between the terminus and upstream areas. The nature of this coupling however is uncertain, since viscous and brittle deformation could come into play.

5. Conclusion

The analysis of high-rate measurements of surface speeds during a large calving event at SKK showed one of the longest calving response lengths so far recorded in Greenland. With an immediate step-wise acceleration detected up to 11 km from the calving front, this event provides evidence for strong coupling between the terminus and upstream areas at short time scales. Higher deformation rates were also detected in the shear margins 3 km from the terminus immediately after the calving event, and occurred more than an order of magnitude faster than during the multi-day glacier acceleration. Our observations therefore suggest that calving events can impact the longer-

term dynamics of the main stream but also the highly-crevassed shear margins, which by themselves are key components in the dynamics of SKK.

Using a simple theoretical framework, this large flow disturbance was used as a full-scale natural experiment to probe SKK's short-term sensitivity to forcings and longer-term force balance. These theoretical considerations showed that the increase of lateral force transfer due to calving was the main process driving the observed response, and suggests that the larger-scale drag provided by the shear margins played a more important role than basal friction.

SKK's dynamic response to calving events is known to have complex links to terminus conditions and over certain periods has no apparent link to calving event magnitude (Cassotto et al., 2019). In addition, the interplay between short-term viscous and brittle deformations, and associated temporal variations, are still poorly constrained. While this complexity can limit the representativeness of isolated events, it also complicates general conclusions from larger catalogs of events. The detailed characterization of dynamic responses to singular large disturbances hence constitutes an important step toward a better understanding of the processes at stake over hours to minutes at fast outlet glaciers.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

TRI data and software associated with the theoretical framework are publicly available on Zenodo (Wehrlé & Rousseau, 2025). The TRI data processing pipelines are also available on Zenodo (M. Lüthi, 2025). MEaSURES product NSIDC-0777 can be downloaded from the NASA NSIDC distributed active archive center (Howat et al., 2022).

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